Title: Lightweight Authentication Protocols: Toward Intrusion Detection on Constrained Devices in Smart Grid Monitoring

Year: 2015

Authors: S. Müller, E. Chen, N. Dubois (Department of Computer Science, Northbridge University)

Keywords: device security, MQTT, smart cities, IoT, digital twins

Abstract:

Intrusion detection on constrained devices remains a critical challenge for wireless sensor networks, particularly in smart grid monitoring. This paper proposes a framework built on lightweight authentication protocols that explicitly accounts for firmware update security. We evaluate the framework on a year-long smart-building trace and compare it against 3 baseline methods. Experimental results show that the proposed approach improves packet delivery ratio by 16.1% while keeping packet delivery ratio within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of wireless sensor networks has changed how researchers and practitioners approach intrusion detection on constrained devices, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, lightweight authentication protocols have attracted sustained attention as a promising route toward solving intrusion detection on constrained devices in realistic settings such as smart grid monitoring. Despite considerable progress, existing solutions for intrusion detection on constrained devices tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in packet delivery ratio frequently come at the expense of energy per transmitted bit, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how lightweight authentication protocols interact with TinyML inference engines when applied to intrusion detection on constrained devices. We make three contributions: a unified formulation of intrusion detection on constrained devices, a practical design built on lightweight authentication protocols, and an empirical evaluation on a year-long smart-building trace. Understanding firmware update security is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Banerjee et al. (2012) introduced an early framework based on lightweight authentication protocols, reporting a 20.4% improvement in energy per transmitted bit over previous baselines. Subsequent work by Ahmed et al. (2014) extended this line of research to smart grid monitoring, although the evaluation was restricted to small-scale settings. A survey by Mukherjee et al. (2008) categorised existing approaches to intrusion detection on constrained devices into three families and highlighted the lack of standard benchmarks. Haddad et al. (2013) analysed firmware update security and showed that purely heuristic solutions cannot guarantee consistent detection rate. More recently, Dubois et al. (2008) combined lightweight authentication protocols with TinyML inference engines, achieving a 39.0% reduction in overhead on a year-long smart-building trace.

Several studies have criticised the reliance on synthetic workloads, arguing that results on a year-long smart-building trace may not generalise to smart grid monitoring. In contrast to these efforts, our work treats intrusion detection on constrained devices and firmware update security jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to intrusion detection on constrained

devices as a persistent obstacle for smart grid monitoring. Despite this body of literature, no prior study has systematically compared lightweight authentication protocols and TinyML inference engines under identical experimental conditions.

3. Methodology

We formalise intrusion detection on constrained devices as a constrained optimisation task in which the objective is to maximise packet delivery ratio while keeping end-to-end delay within a prescribed budget. All experiments were conducted on a year-long smart-building trace, which provides 16 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies lightweight authentication protocols to reduce the effect of firmware update security, followed by a calibration step that stabilises energy per transmitted bit. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 3 baseline methods, including a conventional approach built on TinyML inference engines and two recent variants reported in the literature. The proposed design improves packet delivery ratio by 16.1% relative to the strongest baseline and reduces detection rate by 8.7%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the TinyML inference engines component reduces packet delivery ratio by approximately 4.8%, confirming its contribution. A sensitivity analysis over 32 parameter settings shows that performance remains stable, with variation in packet delivery ratio below 12.8%. Threats to validity include the limited diversity of a year-long smart-building trace and the possibility that deployment in smart grid monitoring introduces conditions not captured in our setup.

4. Conclusion

This paper investigated intrusion detection on constrained devices through the lens of lightweight authentication protocols, and demonstrated that a carefully designed pipeline can improve packet delivery ratio by 16.1% on a year-long smart-building trace. The findings suggest that wireless sensor networks will continue to benefit from closer integration of lightweight authentication protocols and TinyML inference engines, particularly in smart grid monitoring. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of firmware update security. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for intrusion detection on constrained devices.